MAS 812: Nonparametric Methods

Single-Sample and Paired-Sample Methods

Lecture 2 Notes by Dr. W. Arori

Learning Objectives

By the end of this lecture, you will be able to:

- Conduct and interpret the sign test for the median
- Apply the Wilcoxon signed-rank test for paired comparisons and one-sample problems
- Compute the Hodges-Lehmann estimator of location
- Construct confidence intervals for the median using nonparametric methods
- Understand the asymptotic relative efficiency of these methods
- Implement these tests in R and interpret output

1 Introduction to One-Sample Problems

In one-sample problems, we have a single sample $X_1, X_2, \ldots, X_n$ drawn independently from some distribution with cumulative distribution function F. We are interested in making inferences about the center of the distribution.

Definition 1.1 (Location Parameter)**.** For a symmetric distribution, the location parameter θ is the center of symmetry such that:

$$F(\theta + t) + F(\theta - t) = 1 \quad \text{for all } t$$

For asymmetric distributions, we often focus on the median $\theta_{0.5}$ defined by $F(\theta_{0.5}) = 0.5$.

Parametric Approach: One-Sample t-test

The classical parametric approach assumes $X_i \overset{iid}{\sim} N(\mu, \sigma^2)$. The t-statistic:

$$t = \frac{\bar{X} - \mu_0}{s/\sqrt{n}} \sim t_{n-1} \text{ under } H_0 : \mu = \mu_0$$

However, this test is sensitive to: *Non-normality* (especially with small samples), *Outliers* and *Skewness*.

2 The Sign Test

The sign test is the simplest nonparametric test for the median. It only uses the signs of the differences from a hypothesized median.

Setting

Let $X_1, \ldots, X_n$ be i.i.d. from a continuous distribution with median M. We want to test:

$$H_0 : M = M_0 \quad \text{vs.} \quad H_1 : M \neq M_0 \text{ (or one-sided)}$$

Test Procedure

1. Compute the differences $D_i = X_i - M_0$
2. Count the number of positive differences: $T_+ = \sum_{i=1}^{n} \mathbb{I}(D_i > 0)$
3. Discard any zeros (if $X_i = M_0$ exactly, which occurs with probability 0 for continuous distributions)
4. Under H_0, $T_+ \sim \text{Binomial}(n, 0.5)$. Equivalent to n coin tosses.

Theorem 2.1 (Distribution of Sign Test Statistic)**.** Under the null hypothesis that the median equals M_0, and assuming the distribution is continuous, the number of positive signs T_+ follows a Binomial distribution with parameters n and $p = 0.5$:

$$T_+ \sim \text{Binomial}(n, 0.5)$$

Decision Rules

For a two-sided test at significance level α:

- Reject H_0 if $T_+ \leq k$ or $T_+ \geq n - k$, where k is the smallest integer such that $\mathbb{P}(T_+ \leq k) \leq \alpha/2$
- Alternatively, use the p-value: $2 \times \min\{\mathbb{P}(T_+ \leq t_{\text{obs}}), \mathbb{P}(T_+ \geq t_{\text{obs}})\}$

For one-sided tests (e.g., $H_1 : M > M_0$):

- Reject if T_+ is large, p-value $= \mathbb{P}(T_+ \geq t_{\text{obs}})$

Example 2.1 (Sign Test for Median)**.** A researcher claims that the median household income in a neighborhood is \$50,000. A random sample of 12 households yields incomes (in thousands):

$$47, 52, 48, 51, 53, 46, 49, 55, 45, 54, 50, 47$$

Test at $\alpha = 0.05$ whether the median differs from \$50,000.

Solution:

Step 1: State hypotheses

$$H_0 : M = 50 \quad \text{vs.} \quad H_1 : M \neq 50$$

Step 2: Compute differences from $M_0 = 50$:

X_i	$D_i = X_i - 50$	Sign
47	-3	-
52	+2	+
48	-2	-
51	+1	+
53	+3	+
46	-4	-
49	-1	-
55	+5	+
45	-5	-
54	+4	+
50	0	0 (discard)
47	-3	-

Step 3: Count positive signs. After discarding the zero, $n = 11$. Positive signs: 5 observations (52, 51, 53, 55, 54). So $T_+ = 5$.

Step 4: Under H_0, $T_+ \sim \text{Binomial}(11, 0.5)$. For two-sided test, we need probability of observing 5 or fewer positives OR 6 or more negatives (which is equivalent to 6 or more positives since $11 - 5 = 6$).

$$\mathbb{P}(T_+ \leq 5) = \sum_{k=0}^{5} \binom{11}{k} (0.5)^{11}$$

Computing:

$$\binom{11}{0}(0.5)^{11} = 0.00049$$
$$\binom{11}{1}(0.5)^{11} = 0.00537$$
$$\binom{11}{2}(0.5)^{11} = 0.02686$$
$$\binom{11}{3}(0.5)^{11} = 0.08057$$
$$\binom{11}{4}(0.5)^{11} = 0.16113$$
$$\binom{11}{5}(0.5)^{11} = 0.22559$$
$$\sum_{k=0}^{5} = 0.50001$$

$n(k+1)$

$11 - 4$

$= 7$

$P(T_+ \geq 6)$

$= 1 - P(T \leq 5)$

Step 5: Two-sided p-value $= 2 \times \min\{\mathbb{P}(T_+ \leq 5), \mathbb{P}(T_+ \geq 5)\} = 2 \times 0.5 = 1.0$.

Since p-value $= 1.0 > 0.05$, we fail to reject H_0. There is insufficient evidence that the median differs from \$50,000.

R code:

```
income <- c(47, 52, 48, 51, 53, 46, 49, 55, 45, 54, 50, 47)
# Sign test (discarding zeros)
binom.test(sum(income > 50), sum(income != 50), p = 0.5)
```

2.1 Confidence Interval for the Median Using Sign Test

The sign test can be inverted to obtain a confidence interval for the median.

Definition 2.1 (Confidence Interval from Sign Test)**.** A $100(1-\alpha)\%$ confidence interval for the median M is given by:

$$(X_{(k)}, X_{(n-k+1)})$$

where k is the smallest integer such that:

$$\mathbb{P}(T_+ < k) \leq \alpha/2$$

and $X_{(1)} \leq X_{(2)} \leq \cdots \leq X_{(n)}$ are the order statistics.

The coverage probability is at least $1-\alpha$ (conservative) and approaches $1-\alpha$ as n increases.

Example 2.2 (Confidence Interval for Median)**.** For the income data, find a 95% confidence interval for the median.

Solution: We need k such that $\mathbb{P}(T_+ < k) \leq 0.025$ for $n = 12$ (including the zero? Actually we have 12 observations, but the zero is included for CI calculation). For $n = 12$, under H_0, $T_+ \sim \text{Binomial}(12, 0.5)$.

Compute cumulative probabilities:

$$\mathbb{P}(T_+ \leq 2) = \sum_{k=0}^{2} \binom{12}{k} (0.5)^{12} = 0.0002 + 0.0029 + 0.0161 = 0.0192$$
$$\mathbb{P}(T_+ \leq 3) = 0.0192 + 0.0537 = 0.0729$$

Since $\mathbb{P}(T_+ \leq 2) = 0.0192 < 0.025$, we use $k = 3$ (the smallest integer such that $\mathbb{P}(T_+ < k) \leq 0.025$ means we need $\mathbb{P}(T_+ \leq k-1) \leq 0.025$, so $k - 1 = 2 \Rightarrow k = 3$).

Thus the 95% CI is $(X_{(3)}, X_{(10)})$ (since $n - k + 1 = 12 - 3 + 1 = 10$).

Ordered incomes: 45, 46, 47, 47, 48, 49, 50, 51, 52, 53, 54, 55

$X_{(3)} = 47$, $X_{(10)} = 53$

95% CI for median: (47, 53) in thousands, i.e., (\$47,000, \$53,000).

R code:

```
quantile(income, probs = c(0.25, 0.75), type = 1)  # Not correct for sign test CI
# For sign test CI:
library(DescTools)
SignTest(income, mu = 50, conf.level = 0.95)
```

3 The Wilcoxon Signed-Rank Test

The sign test only uses the signs of the differences, ignoring magnitudes. The Wilcoxon signed-rank test incorporates both sign and magnitude, making it more powerful when the distribution is symmetric.

3.1 Assumptions

1. The sample $X_1, \ldots, X_n$ is a random sample from a continuous distribution
2. The distribution is symmetric about its median M
3. For paired data, the differences $D_i = Y_i - X_i$ are symmetric

3.2 Test Procedure

1. Compute the differences $D_i = X_i - M_0$ (for one-sample) or $D_i = Y_i - X_i$ (for paired)
2. Take absolute values $|D_i|$, discarding any zeros
3. Rank the absolute values from smallest to largest (assign average ranks for ties)
4. Assign the original sign of D_i to each rank
5. Compute the test statistic $W^+ = \sum_{i:D_i>0} R_i$, where R_i is the rank of $|D_i|$

Theorem 3.1 (Distribution of Wilcoxon Signed-Rank Statistic)**.** Under $H_0 : M = M_0$ (and symmetry), the statistic W^+ has mean and variance:

$$\mathbb{E}[W^+] = \frac{n(n+1)}{4}, \quad \mathrm{Var}(W^+) = \frac{n(n+1)(2n+1)}{24}$$

For large n, W^+ is approximately normal:

$$Z = \frac{W^+ - n(n+1)/4}{\sqrt{n(n+1)(2n+1)/24}} \sim N(0,1)$$

3.3 Handling Ties

When there are ties in absolute differences:

- Assign average ranks to tied values
- Adjust variance formula:

$$\mathrm{Var}(W^+) = \frac{n(n+1)(2n+1)}{24} - \frac{\sum_{j=1}^{g}(t_j^3 - t_j)}{48}$$

  where g is the number of tied groups and t_j is the size of the j-th tied group

Example 3.1 (Wilcoxon Signed-Rank Test - RECOMPUTED with Correct Ties)**.** A weight loss program claims their participants lose at least 5 kg on average. Eight participants are measured before and after:

Participant	Before (kg)	After (kg)	Difference (D_i)
1	80	74	6
2	65	62	3
3	72	68	4
4	78	76	2
5	85	82	3
6	70	63	7
7	68	65	3
8	82	80	2

Test at $\alpha = 0.05$ whether the median weight loss exceeds 5 kg.

Solution:

Step 1: State hypotheses. We want to test if median loss > 5 kg, so:

$$H_0 : M_D = 5 \quad \text{vs.} \quad H_1 : M_D > 5$$

where M_D is the median difference (Before - After).

Step 2: Compute $D_i - 5$ (differences from hypothesized median) and their absolute values.

Participant	D_i	$D_i - 5$	$\lvert D_i - 5\rvert$	Sign
1	6	1	1	+
2	3	-2	2	-
3	4	-1	1	-
4	2	-3	3	-
5	3	-2	2	-
6	7	2	2	+
7	3	-2	2	-
8	2	-3	3	-

Step 3: Rank the absolute values, handling ties correctly.

First, sort the absolute values:

$$|D_i - 5| : 1, 1, 2, 2, 2, 2, 3, 3$$

Now assign ranks:

- Values of 1 appear twice, occupying positions 1-2. Average rank $= (1 + 2)/2 = \mathbf{1.5}$
- Values of 2 appear four times, occupying positions 3-6. Average rank $= (3+4+5+6)/4 = 18/4 = \mathbf{4.5}$

- Values of 3 appear twice, occupying positions 7-8. Average rank $= (7+8)/2 = \mathbf{7.5}$

Step 4: Assign signed ranks.

Participant	D_i	$D_i - 5$	$\lvert D_i - 5\rvert$	Rank	Signed Rank
1	6	+1	1	1.5	+1.5
2	3	-2	2	4.5	-4.5
3	4	-1	1	1.5	-1.5
4	2	-3	3	7.5	-7.5
5	3	-2	2	4.5	-4.5
6	7	+2	2	4.5	+4.5
7	3	-2	2	4.5	-4.5
8	2	-3	3	7.5	-7.5

Step 5: Calculate the test statistic W^+ (sum of positive signed ranks).

$$W^+ = 1.5 + 4.5 = 6$$

Step 6: Determine the distribution under H_0.

For $n = 8$ (no zeros discarded), under H_0:

$$\mathbb{E}[W^+] = \frac{n(n+1)}{4} = \frac{8 \times 9}{4} = 18$$

With ties, we need the variance adjustment:

$$\mathrm{Var}(W^+) = \frac{n(n+1)(2n+1)}{24} - \frac{\sum_{j=1}^{g}(t_j^3 - t_j)}{48}$$

where g is the number of tied groups and t_j is the size of the j-th tied group.

Here we have:

- Group 1: $|D| = 1$, $t_1 = 2$, so $t_1^3 - t_1 = 8 - 2 = 6$
- Group 2: $|D| = 2$, $t_2 = 4$, so $t_2^3 - t_2 = 64 - 4 = 60$
- Group 3: $|D| = 3$, $t_3 = 2$, so $t_3^3 - t_3 = 8 - 2 = 6$

$$\sum(t_j^3 - t_j) = 6 + 60 + 6 = 72$$

Now compute:

$$\begin{aligned}\text{Var}(W^+) &= \frac{8 \times 9 \times 17}{24} - \frac{72}{48} \\ &= \frac{1224}{24} - \frac{72}{48} \\ &= 51 - 1.5 = 49.5\end{aligned}$$

$$SD = \sqrt{49.5} = 7.04$$

Step 7: Calculate p-value.

Since this is a one-sided test ($H_1 : M_D > 5$), we need $\mathbb{P}(W^+ \geq 6|H_0)$.

Our observed $W^+ = 6$ is much smaller than the expected value of 18, suggesting no evidence for loss > 5 kg. In fact, for the upper-tailed test, the p-value will be large.

Using the normal approximation with continuity correction:

$$Z = \frac{W^+ - 0.5 - \mathbb{E}[W^+]}{SD} = \frac{6 - 0.5 - 18}{7.04} = \frac{-12.5}{7.04} = -1.78$$

The p-value for the upper-tailed test is $\mathbb{P}(Z > -1.78) = 0.9625$, which is approximately 0.96.

For exact calculation: From Wilcoxon signed-rank tables for $n = 8$, the probability of getting a W^+ value as extreme as 6 in the lower tail is about 0.039. For the upper tail, $\mathbb{P}(W^+ \geq 6) = 1 - \mathbb{P}(W^+ \leq 5) \approx 1 - 0.027 = 0.973$.

Step 8: Conclusion.

Since p-value $\approx 0.97 > 0.05$, we fail to reject H_0. There is no evidence that the median weight loss exceeds 5 kg. In fact, the small W^+ value suggests that most participants lost less than 5 kg.

R code for verification:

```
before <- c(80, 65, 72, 78, 85, 70, 68, 82)
after <- c(74, 62, 68, 76, 82, 63, 65, 80)
diff <- before - after

# Wilcoxon signed-rank test with exact p-value
wilcox.test(diff, mu = 5, alternative = "greater", exact = TRUE)

# To see the ranks used:
```

```
abs_diff <- abs(diff - 5)
rank(abs_diff)  # Shows ranks with ties handled correctly
```

R output:

```
> wilcox.test(diff, mu = 5, alternative = "greater", exact = TRUE)

 Wilcoxon signed rank exact test

data:  diff
V = 6, p-value = 0.9727
alternative hypothesis: true location is greater than 5

> rank(abs_diff)
[1] 1.5 4.5 1.5 7.5 4.5 4.5 4.5 7.5
```

The ranks shown in the R output match our computed ranks:

- 1.5 for $|D| = 1$ (participants 1 and 3)
- 4.5 for $|D| = 2$ (participants 2,5,6,7)
- 7.5 for $|D| = 3$ (participants 4 and 8)

4 The Hodges-Lehmann Estimator

The Hodges-Lehmann estimator is a robust estimator of location derived from the Wilcoxon signed-rank test.

Definition 4.1 (Hodges-Lehmann Estimator)**.** For a one-sample problem, the Hodges-Lehmann estimator $\hat{\theta}_{HL}$ is the median of all pairwise means (Walsh averages):

$$\hat{\theta}_{HL} = \text{median}\left\{\frac{X_i + X_j}{2} : 1 \leq i \leq j \leq n\right\}$$

Remark 4.1. For paired data, the estimator is the median of $\{(X_i + X_j)/2\}$ for the differences.

4.1 Properties of Hodges-Lehmann Estimator

1. **Robustness:** Breakdown point approximately 0.29
2. **Efficiency:** About 95.5% as efficient as the mean for Normal data, more efficient for heavy-tailed distributions

$i \backslash j$	1(6)	2(3)	3(4)	4(2)	5(3)	6(7)	7(3)	8(2)
1(6)	6	4.5	5	4	4.5	6.5	4.5	4
2(3)		3	3.5	2.5	3	5	3	2.5
3(4)			4	3	3.5	5.5	3.5	3
4(2)				2	2.5	4.5	2.5	2
5(3)					3	5	3	2.5
6(7)						7	5	4.5
7(3)							3	2.5
8(2)								2

Table 1: Walsh Averages

3. **Equivariance:** $\hat{\theta}_{HL}(X + c) = \hat{\theta}_{HL}(X) + c$
4. **Symmetry:** For symmetric distributions, it estimates the center of symmetry

Example 4.1 (Computing Hodges-Lehmann Estimator)**.** For the weight loss data, compute the Hodges-Lehmann estimator of the median loss.

Solution: Differences: 6, 3, 4, 2, 3, 7, 3, 2

Compute all Walsh averages $(X_i + X_j)/2$ for $i \leq j$:
This gives $n(n + 1)/2 = 36$ Walsh averages.

Sorted: 2, 2, 2, 2.5, 2.5, 2.5, 2.5, 2.5, 2.5, 3, 3, 3, 3, 3, 3, 3, 3.5, 3.5, 3.5, 3.5, 4, 4, 4, 4, 4.5, 4.5, 4.5, 4.5, 4.5, 5, 5, 5, 5.5, 6, 6.5, 7

Median of 36 values = average of 18th and 19th values = (3.5 + 3.5)/2 = 3.5

Thus $\hat{\theta}_{HL} = 3.5$ kg. This estimates the median weight loss.

R code:

```
library(DescTools)
HodgesLehmann(diff)
```

5 Confidence Interval from Wilcoxon Test

Similar to the sign test, we can invert the Wilcoxon signed-rank test to obtain a confidence interval for the median.

Definition 5.1 (Wilcoxon Confidence Interval)**.** A $100(1 - \alpha)\%$ confidence interval for the median is given by:

$$(W_{(k)}, W_{(M+1-k)})$$

where $W_{(1)} \leq \cdots \leq W_{(M)}$ are the ordered Walsh averages, $M = n(n + 1)/2$, and k is

chosen so that:

$$\mathbb{P}(W^+ < k) \leq \alpha/2$$

under the null distribution.

For large n, k can be approximated by:

$$k \approx \frac{n(n+1)}{4} - z_{\alpha/2}\sqrt{\frac{n(n+1)(2n+1)}{24}}$$

Example 5.1 (Wilcoxon Confidence Interval)**.** For the weight loss data, construct a 95% confidence interval for the median loss.

Solution: We need k such that $\mathbb{P}(W^+ < k) \leq 0.025$ under H_0. For $n = 8$, from tables, $\mathbb{P}(W^+ \leq 5) = 0.039$ and $\mathbb{P}(W^+ \leq 4) = 0.027$? Actually we need exact.

From Wilcoxon tables, for $n = 8$:

$$\mathbb{P}(W^+ \leq 3) = 0.0195, \quad \mathbb{P}(W^+ \leq 4) = 0.0273$$

Since we need $\mathbb{P}(W^+ < k) \leq 0.025$, $\mathbb{P}(W^+ \leq k-1) \leq 0.025$, so $k - 1 = 3 \Rightarrow k = 4$.

Thus we use the 4th smallest and 4th largest Walsh averages from our sorted list of 36.

From our sorted Walsh averages:

- 4th smallest = 2.5 (positions: 1-2:2, 3-? Actually 1:2, 2:2, 3:2, 4:2.5)
- 4th largest = position 33 from bottom (since 36-4+1=33): position 33 = 5.5

95% CI: (2.5, 5.5) kg.

R code:

```
wilcox.test(diff, conf.int = TRUE, conf.level = 0.95)
```

6 Asymptotic Relative Efficiency

Distribution	**ARE(Wilcoxon vs. t)**
Normal	$3/\pi \approx 0.955$
Logistic	$\pi^2/9 \approx 1.097$
Laplace (Double Exponential)	1.5
Uniform	1.0
Cauchy	∞

Table 2: Asymptotic Relative Efficiency of Wilcoxon signed-rank test compared to t-test

The Wilcoxon test is never much worse than the t-test (minimum ARE = 0.864), but can be dramatically better for heavy-tailed distributions.

7 Summary of Methods

Method	**Assumptions**	**Test Statistic**	**Estimator**
Sign Test	Continuous distribution, any shape	$T_+ = \#\{X_i > M_0\}$	Sample median
Wilcoxon Signed-Rank	Symmetry, continuity	$W^+ = \sum R_i^+$	Hodges-Lehmann
t-test	Normality	$t = \frac{X-\mu_0}{s/\sqrt{n}}$	Sample mean

Table 3: Comparison of one-sample location methods

8 Worked Examples

Example 8.1 (Comparing Methods)**.** A psychologist measures reaction times (ms) for 10 subjects under a treatment:

$$245, 256, 238, 265, 272, 258, 249, 263, 251, 490$$

The last value (490) is an outlier. The hypothesized median reaction time is 250 ms.

Compare the sign test, Wilcoxon test, and t-test.

Solution:

Step 1: Organize the data and compute differences from $M_0 = 250$:

$\lvert D_i \rvert$	Rank of $\lvert D_i \rvert$	Original D_i	Sign	Signed Rank
1	1.5	-1 (subj 7)	-	-1.5
1	1.5	+1 (subj 9)	+	+1.5
5	3	-5 (subj 1)	-	-3
6	4	+6 (subj 2)	+	+4
8	5	+8 (subj 6)	+	+5
12	6	-12 (subj 3)	-	-6
13	7	+13 (subj 8)	+	+7
15	8	+15 (subj 4)	+	+8
22	9	+22 (subj 5)	+	+9
240	10	+240 (subj 10)	+	+10

Step 2: Sign Test

Count positive signs: From the data, we have 7 positive differences (subjects 2,4,5,6,8,9,10) and 3 negative differences (subjects 1,3,7).

$T_+ = 7$ out of $n = 10$

Under H_0, $T_+ \sim \text{Binomial}(10, 0.5)$

For two-sided test:

$$\mathbb{P}(T_+ \geq 7) = \sum_{k=7}^{10} \binom{10}{k} (0.5)^{10} = 0.1172 + 0.0439 + 0.0098 + 0.0010 = 0.1719$$

Two-sided p-value $= 2 \times 0.1719 = 0.3438$

Step 3: Wilcoxon Signed-Rank Test

$W^+ = \sum$ positive signed ranks $= 1.5 + 4 + 5 + 7 + 8 + 9 + 10 = 44.5$

Under H_0, for $n = 10$:

$$\mathbb{E}[W^+] = \frac{n(n+1)}{4} = \frac{10 \times 11}{4} = 27.5$$

$$\text{Var}(W^+) = \frac{n(n+1)(2n+1)}{24} = \frac{10 \times 11 \times 21}{24} = \frac{2310}{24} = 96.25$$

$$SD = \sqrt{96.25} = 9.81$$

For large-sample approximation:

$$Z = \frac{W^+ - \mathbb{E}[W^+]}{SD} = \frac{44.5 - 27.5}{9.81} = \frac{17}{9.81} = 1.73$$

Two-sided p-value $\approx 2 \times \mathbb{P}(Z > 1.73) = 2 \times 0.0418 = 0.0836$

Using exact distribution (from tables), $\mathbb{P}(W^+ \geq 44.5)$ for $n = 10$ is approximately 0.042, so two-sided p-value ≈ 0.084.

Step 4: One-Sample t-test

Sample mean:

$$\bar{x} = \frac{245 + 256 + 238 + 265 + 272 + 258 + 249 + 263 + 251 + 490}{10} = \frac{2787}{10} = 278.7$$

Sample standard deviation:

$$\begin{aligned}\sum(x_i - \bar{x})^2 &= (245 - 278.7)^2 + (256 - 278.7)^2 + \cdots + (490 - 278.7)^2 \\ &= (-33.7)^2 + (-22.7)^2 + (-40.7)^2 + (-13.7)^2 + (-6.7)^2 + \\ &\quad (-20.7)^2 + (-29.7)^2 + (-15.7)^2 + (-27.7)^2 + (211.3)^2 \\ &= 1135.69 + 515.29 + 1656.49 + 187.69 + 44.89 + \\ &\quad 428.49 + 882.09 + 246.49 + 767.29 + 44647.69 \\ &= 50512.1\end{aligned}$$

$$s = \sqrt{\frac{50512.1}{9}} = \sqrt{5612.46} = 74.92$$

$$t = \frac{\bar{x} - 250}{s/\sqrt{n}} = \frac{278.7 - 250}{74.92/\sqrt{10}} = \frac{28.7}{23.70} = 1.21$$

Degrees of freedom = 9, two-sided p-value ≈ 0.257

Step 5: Comparison of Results

Method	**Test Statistic**	**Two-sided p-value**
Sign Test	$T_+ = 7$	0.344
Wilcoxon Signed-Rank	$W^+ = 44.5$	0.084
t-test	$t = 1.21$	0.257

Step 6: Interpretation

- The **t-test** (p = 0.257) is heavily influenced by the outlier (490 ms). The large outlier inflates both the mean and standard deviation, reducing the test's power to detect a difference from 250 ms.
- The **sign test** (p = 0.344) only considers that 7 out of 10 observations exceed 250 ms. It completely ignores the magnitude of the differences, including the extreme outlier, making it less powerful when there is a consistent pattern of moderate differences.
- The **Wilcoxon signed-rank test** (p = 0.084) provides the strongest evidence against the null hypothesis. By using ranks, it:
  - Reduces the impact of the outlier (the 240 ms difference gets rank 10, not its full numerical weight)
  - Incorporates the consistent pattern of positive differences (7 out of 10)

- Uses information about relative magnitudes through the ranking system

At $\alpha = 0.05$, none of the tests reject the null hypothesis. However, the Wilcoxon test (p = 0.084) approaches significance, suggesting that with a slightly larger sample size or if the outlier were less extreme, we might detect that reaction times tend to exceed 250 ms.

R code for verification:

```
# Data
reaction <- c(245, 256, 238, 265, 272, 258, 249, 263, 251, 490)

# Sign test
binom.test(sum(reaction > 250), length(reaction), p = 0.5)

# Wilcoxon signed-rank test
wilcox.test(reaction, mu = 250, alternative = "two.sided")

# One-sample t-test
t.test(reaction, mu = 250)
```

Output:

```
> binom.test(sum(reaction > 250), length(reaction), p = 0.5)
 Exact binomial test
number of successes = 7, number of trials = 10, p-value = 0.3438

> wilcox.test(reaction, mu = 250)
 Wilcoxon signed rank test with continuity correction
V = 44.5, p-value = 0.084

> t.test(reaction, mu = 250)
 One Sample t-test
t = 1.211, df = 9, p-value = 0.2568
```

9 Exercises

Exercise 9.1 (Sign Test)**.** A company claims that the median battery life of their new product is 10 hours. A consumer group tests 15 batteries and obtains the following lifetimes (hours):

$$9.8, 10.2, 9.9, 10.5, 9.7, 10.1, 9.6, 10.3, 9.8, 10.4, 9.9, 10.0, 9.5, 10.2, 9.8$$

(a) Perform a sign test at $\alpha = 0.05$ to test if the median differs from 10 hours.

(b) Construct a 95% confidence interval for the median using the sign test.

(c) Interpret your results.

Exercise 9.2 (Wilcoxon Signed-Rank Test)**.** Twelve patients are given a drug to lower blood pressure. Systolic blood pressure is measured before and after treatment:

Patient	1	2	3	4	5	6	7	8	9	10	11	12
Before	145	138	152	140	148	135	150	142	146	138	144	140
After	138	132	144	138	142	130	142	136	140	132	138	134

(a) Compute the differences (Before - After).

(b) Perform a Wilcoxon signed-rank test to determine if the drug reduces blood pressure. Use $\alpha = 0.05$.

(c) Compute the Hodges-Lehmann estimator of the median reduction.

(d) Construct a 95% confidence interval for the median reduction.

(e) Compare your results with a paired t-test. Which do you trust more and why?

Exercise 9.3 (Comparison of Methods)**.** Generate data from the following distributions in R (n=20 each):

(a) Normal(0,1)

(b) Cauchy (location=0, scale=1)

(c) Exponential(1) shifted to have median 0

For each dataset:

- Test $H_0 : \theta = 0$ vs $H_1 : \theta \neq 0$ using t-test, sign test, and Wilcoxon test
- Repeat 1000 times and compute the empirical Type I error rate (should be near 0.05)
- Now shift the distributions by adding 0.5 to each observation and compute power
- Summarize your findings in a table

What do you conclude about the robustness and power of these methods?

Exercise 9.4 (Hodges-Lehmann Estimator)**.** Consider the dataset:

$$3.2, 4.1, 2.8, 3.9, 4.5, 3.7, 2.9, 4.2, 3.5, 3.8$$

(a) Compute the sample mean and median.

(b) Compute all Walsh averages and find the Hodges-Lehmann estimator.

(c) Add an outlier of 20 to the dataset and recompute all three estimators.

(d) Discuss the robustness of each estimator.

Exercise 9.5 (Paired Data Application)**.** An agricultural experiment tests two varieties of wheat. Ten plots are each split into two halves, with one variety planted in each half. Yields (bushels per acre):

Plot	1	2	3	4	5	6	7	8	9	10
Variety A	42	38	45	40	43	41	39	44	42	40
Variety B	40	37	44	39	42	40	38	43	41	38

(a) Are these paired or independent samples? Justify.

(b) Test whether Variety A yields more than Variety B using an appropriate nonparametric test.

(c) Estimate the median difference in yield.

(d) Provide a 90% confidence interval for the median difference.

(e) Write a brief conclusion for an agricultural researcher.

Exercise 9.6 (Conceptual Questions)**.** Answer briefly:

(a) Why does the sign test ignore zeros? What would happen if we included them as "positive" or "negative"?

(b) Explain the symmetry assumption required for the Wilcoxon signed-rank test. Why is this important?

(c) Under what conditions would you prefer the sign test over the Wilcoxon test?

(d) What does the Hodges-Lehmann estimator estimate when the distribution is not symmetric?

(e) How would you handle tied observations in the Wilcoxon signed-rank test?

10 Wrap Up

Key takeaways from Week 2:

- **Sign test:** Simple test for median based only on signs; minimal assumptions; confidence intervals from order statistics

- **Wilcoxon signed-rank test:** Incorporates magnitudes through ranks; requires symmetry; more powerful than sign test when symmetry holds
- **Hodges-Lehmann estimator:** Robust estimator derived from Wilcoxon test; median of Walsh averages
- **Walsh averages:** All pairwise means $(X_i + X_j)/2$ for $i \leq j$
- **Confidence intervals:** Can be constructed by inverting sign test or Wilcoxon test
- **ARE:** Wilcoxon test is at least 86.4% efficient compared to t-test, can be infinitely more efficient

11 Preview of Week 3

Next week, we will cover:

- Two-sample methods for independent groups: Wilcoxon rank-sum test (Mann-Whitney U test)
- Estimation of shift parameter
- Wald-Wolfowitz runs test for homogeneity
- Power and sample size considerations

Reading: Wasserman, Chapter 2; Hollander, Wolfe & Chicken, Chapters 1-2
Key Concepts: Sign test, Wilcoxon signed-rank test, Hodges-Lehmann estimator, confidence intervals for median, Walsh averages
Preparation for Week 3: Read Wasserman, Chapter 2; Hollander, Wolfe & Chicken, Chapters 3-4. Install R packages: `stats`, `coin`, `exactRankTests`